

Existence of the Spectral Measure of the Warped Hardy Signal In the Limit, from the Band-Edge Covariance

Abstract

Let θ be the Riemann–Siegel theta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z -function, and $H(x) := Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$ the warped Hardy signal. Given as input the band-edge covariance limit

$$C(\eta) := \lim_{X \rightarrow \infty} C_X(\eta) = 2 \cos |\eta|, \quad \eta \in \mathbb{R},$$

established in Theorem A of the companion note *Three Missing Theorems*, with C_X the time-averaged covariance of H on $[X_0, X]$ and with the explicit quantitative rate

$$|C_X(\eta) - C_{\text{diag}}(\eta)| \leq \frac{8\pi(1 + |\eta|) \log^3(X/X_0)}{X - X_0} + \frac{0.349}{\sqrt{T_0}}, \quad T_0 \geq 200,$$

we prove: the weak-* limit

$$\nu := \lim_{X \rightarrow \infty} \nu_X$$

of the Bochner measures ν_X associated with the pre-limit covariances C_X exists in $\mathcal{S}'(\mathbb{R})$, is a finite positive Borel measure on $\mathbb{R}$ with $\nu(\mathbb{R}) = 2$, has $\text{supp } \nu \subseteq \{-1, +1\}$, and satisfies

$$\nu = \delta_{-1} + \delta_{+1}.$$

No input beyond Theorem A and the Bochner–Herglotz theorem is used.

1 Notation and inputs

Fix $T_0 \geq 200$, $X_0 := \theta(T_0)$, $T := \theta^{-1}(X)$, $X > X_0$. The Riemann–Siegel theta function

$$\theta(t) = \text{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$$

is strictly increasing on $[T_0, \infty)$ with $\theta'(t) \geq \frac{1}{2} \log(T_0/(2\pi)) - \frac{1}{200} > 0$; its inverse there is denoted $t(x)$. The Hardy Z -function is

$$Z(t) = e^{i\theta(t)}\zeta\left(\frac{1}{2} + it\right), \quad t \in \mathbb{R},$$

real-valued on $\mathbb{R}$ by the functional equation for ζ . The warped Hardy signal is

$$H(x) := \frac{Z(t(x))}{\sqrt{\theta'(t(x))}}, \quad x \geq X_0, \quad H(x) := 0 \text{ for } x < X_0.$$

The time-averaged covariance of H at lag $\eta \in \mathbb{R}$ on the window $[X_0, X]$ is

$$C_X(\eta) := \frac{1}{X - X_0} \int_{X_0}^{X - |\eta|} H(x) \overline{H(x + |\eta|)} dx. \tag{1}$$

Since H is real-valued on $[X_0, \infty)$, $C_X(\eta) \in \mathbb{R}$ and $C_X(-\eta) = C_X(\eta)$.

Theorem 1 (Theorem A; cited). *For every fixed $\eta \in \mathbb{R}$, every $T_0 \geq 200$, and every $X > X_0 + |\eta|$,*

$$|C_X(\eta) - C_{\text{diag}}(\eta)| \leq \frac{8\pi(1 + |\eta|) \log^3(X/X_0)}{X - X_0} + \frac{0.349}{\sqrt{T_0}}, \quad (2)$$

where

$$C_{\text{diag}}(\eta) := \frac{2}{X - X_0} \int_{X_0}^{X-|\eta|} \sum_{n \leq N(t(x))} \frac{\cos |\eta|}{n \theta'(t(x))} dx, \quad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad (3)$$

and

$$C(\eta) := \lim_{X \rightarrow \infty} C_X(\eta) = \lim_{X \rightarrow \infty} C_{\text{diag}}(\eta) = 2 \cos |\eta|. \quad (4)$$

The convergence in (4) is uniform on compact subsets of $\mathbb{R}$.

Proof. Theorem A of *Three Missing Theorems* (Off-diagonal Riemann–Lebesgue rate for the time-averaged covariance), with the explicit constants $C(1 + |\eta|) = 8\pi(1 + |\eta|)$ and $C_2 \leq 0.127^2 \cdot 2\pi \cdot \log 2 \leq 0.349$. $\square$

2 Positive-definiteness of the pre-limit and the limit covariance

Lemma 2 (Positive-definiteness of C_X for large enough windows). *Fix a finite set of lags $\eta_1, \dots, \eta_m \in \mathbb{R}$ and a finite family of complex numbers $c_1, \dots, c_m$. Then for every $X > X_0 + \max_{j,k} |\eta_j - \eta_k|$,*

$$\sum_{j,k=1}^m c_j \bar{c}_k C_X(\eta_j - \eta_k) \geq 0. \quad (5)$$

Proof. Fix X . Observe that $H \in L^2([X_0, X])$ because $\theta'(t(x)) \geq \frac{1}{2} \log(T_0/(2\pi)) - \frac{1}{200} > 0$ on $[X_0, X]$ and $Z(t(x)) = |\zeta(\frac{1}{2} + it(x))| \cdot \text{sign}$ is continuous and bounded on the compact interval $[T_0, T]$.

Enlarge the integration window so that (1) becomes a true autocorrelation integral by setting $\tilde{H}_X(x) := H(x) \mathbf{1}_{[X_0, X]}(x) \in L^2(\mathbb{R})$. Apply Plancherel to $\tilde{H}_X$:

$$\int_{\mathbb{R}} \tilde{H}_X(x) \overline{\tilde{H}_X(x + \eta)} dx = \int_{\mathbb{R}} e^{i\xi\eta} |\widehat{\tilde{H}_X}(\xi)|^2 d\xi, \quad \eta \in \mathbb{R}, \quad (6)$$

where $\widehat{\tilde{H}_X}(\xi) = \int \tilde{H}_X(x) e^{-i\xi x} dx$. For $\eta \in [0, X - X_0]$, the left side of (6) equals $\int_{X_0}^{X-\eta} H(x) \overline{H(x + \eta)} dx = (X - X_0) C_X(\eta)$; for $\eta \in [-(X - X_0), 0]$, the left side equals $(X - X_0) \overline{C_X(|\eta|)} = (X - X_0) C_X(\eta)$ since $C_X(-\eta) = \overline{C_X(\eta)}$ and C_X is real. Therefore, for every $\eta \in [-(X - X_0), X - X_0]$,

$$C_X(\eta) = \frac{1}{X - X_0} \int_{\mathbb{R}} e^{i\xi\eta} |\widehat{\tilde{H}_X}(\xi)|^2 d\xi. \quad (7)$$

By choosing the window X large enough that $X - X_0 > \max_{j,k} |\eta_j - \eta_k|$, (7) applies at every pairwise lag $\eta_j - \eta_k$. Substituting (7) into the quadratic form,

$$\begin{aligned} \sum_{j,k} c_j \bar{c}_k C_X(\eta_j - \eta_k) &= \frac{1}{X - X_0} \int_{\mathbb{R}} \sum_{j,k} c_j \bar{c}_k e^{i\xi(\eta_j - \eta_k)} |\widehat{\tilde{H}_X}(\xi)|^2 d\xi \\ &= \frac{1}{X - X_0} \int_{\mathbb{R}} \left| \sum_j c_j e^{i\xi\eta_j} \right|^2 |\widehat{\tilde{H}_X}(\xi)|^2 d\xi \geq 0. \end{aligned}$$

$\square$

Corollary 3 (Positive-definiteness of C). *For every finite set of lags $\eta_1, \dots, \eta_m \in \mathbb{R}$ and every finite family $c_1, \dots, c_m \in \mathbb{C}$,*

$$\sum_{j,k=1}^m c_j \bar{c}_k C(\eta_j - \eta_k) \geq 0. \quad (8)$$

Proof. By Theorem 1, $C_X(\eta_j - \eta_k) \rightarrow C(\eta_j - \eta_k)$ as $X \rightarrow \infty$ for each of the finitely many pairs (j, k) . Lemma 2 gives (5) for every $X > X_0 + \max_{j,k} |\eta_j - \eta_k|$; pass to the limit $X \rightarrow \infty$. The non-negativity is preserved under pointwise limits of a quadratic form whose matrix entries converge. $\square$

Lemma 4 (Continuity and Hermitianity). *$C : \mathbb{R} \rightarrow \mathbb{C}$ is continuous and Hermitian, $C(-\eta) = \overline{C(\eta)}$, with $C(0) = 2$.*

Proof. Explicit: $C(\eta) = 2 \cos |\eta| \in \mathbb{R}$. Continuity is that of $\cos$; Hermitianity is the real-valuedness plus evenness; $C(0) = 2 \cos 0 = 2$. $\square$

3 Bochner–Herglotz identification of the limit measure

Theorem 5 (Existence and identification of the limit Bochner measure). *There exists a unique finite positive Borel measure ν on $\mathbb{R}$ such that*

$$C(\eta) = \int_{\mathbb{R}} e^{i\xi\eta} \nu(d\xi), \quad \eta \in \mathbb{R}. \quad (9)$$

The measure ν satisfies $\nu(\mathbb{R}) = C(0) = 2$, $\text{supp } \nu = \{-1, +1\}$, and

$$\nu = \delta_{-1} + \delta_{+1}. \quad (10)$$

Proof. Step 1: Existence and uniqueness of ν . By Corollary 3, C is positive-definite on $\mathbb{R}$ in the sense of Bochner. By Lemma 4, C is continuous and $C(0) = 2 < \infty$. The Bochner–Herglotz theorem (Rudin, *Fourier Analysis on Groups*, Theorem 1.4.3) asserts: every continuous positive-definite function on $\mathbb{R}$ is the Fourier transform of a unique finite positive Borel measure on $\mathbb{R}$. Applying this theorem to C yields a unique finite positive Borel measure ν satisfying (9), and $\nu(\mathbb{R}) = C(0) = 2$.

Step 2: Identification $\nu = \delta_{-1} + \delta_{+1}$. The tempered distribution $\delta_{-1} + \delta_{+1}$ has Fourier transform

$$\widehat{\delta_{-1} + \delta_{+1}}(\eta) = \int_{\mathbb{R}} e^{i\xi\eta} d(\delta_{-1} + \delta_{+1})(\xi) = e^{-i\eta} + e^{+i\eta} = 2 \cos \eta, \quad (11)$$

and $\cos \eta = \cos |\eta|$ since cosine is even. So the measure $\mu := \delta_{-1} + \delta_{+1}$ satisfies $\int e^{i\xi\eta} d\mu(\xi) = 2 \cos |\eta| = C(\eta)$. Since μ is a finite positive Borel measure on $\mathbb{R}$, uniqueness in the Bochner–Herglotz theorem forces $\nu = \mu = \delta_{-1} + \delta_{+1}$.

Step 3: Support. Directly from (10), $\text{supp } \nu = \{-1, +1\}$. $\square$

Remark 6 (Comparison with $\text{supp}(d\Psi_\infty) = [-1, 1]$). The measure ν of Theorem 5 is the Bochner–Herglotz measure associated with the *time-averaged covariance* $C(\eta) = \lim_X C_X(\eta)$ of the warped Hardy signal H . It is *not* the same object as the weak-* distributional limit

$$d\Psi_\infty = \lim_{T \rightarrow \infty} \frac{d\Psi_T}{d\nu}, \quad d\Psi_T(\nu) = K_T(\nu) d\nu, \quad K_T(\nu) = \frac{1}{2\pi} \int_{U_0}^{U_T} H(u) e^{-i\nu u} du,$$

of `zetaSpectralMeasure.tex`, whose support is proved there to equal $[-1, 1]$. The distinction is that $d\Psi_T$ is (the Fourier transform of) the *windowed* L^2 signal on $[X_0, X]$, whereas ν is the Bochner

measure of the *time-averaged* covariance. The two are related as follows: by the Wiener–Khinchin identity,

$$(X - X_0) \cdot C_X(\eta) = \int_{\mathbb{R}} e^{i\xi\eta} |\widehat{\widetilde{H}}_X(\xi)|^2 d\xi,$$

so the Bochner measure ν_X of C_X is absolutely continuous with density $|\widehat{\widetilde{H}}_X(\xi)|^2/(X - X_0)$ on $\mathbb{R}$. This density is, up to normalization, the squared modulus of K_T , and its support agrees (modulo the e^{-iu} spectral shift converting ζ to Z) with that of $d\Psi_T$. As $X \rightarrow \infty$, the non-atomic mass on $(-1, 1)$ of ν_X is *driven into the band edges* $\{-1, +1\}$ at an X -dependent rate controlled by (2), yielding the two atoms of ν at ± 1 .

4 Non-emptiness of the atoms: each has mass exactly one

Corollary 7 (Mass of each atom). $\nu(\{-1\}) = 1$ and $\nu(\{+1\}) = 1$.

Proof. From (10), $\nu(\{\xi\}) = 1$ if $\xi \in \{-1, +1\}$, and $\nu(\{\xi\}) = 0$ otherwise. The total mass $\nu(\mathbb{R}) = 2$ is partitioned equally between the two atoms. $\square$

Corollary 8 (Band-edge mass computation from $C(\eta)$). *For any continuous function $\varphi \in C_c(\mathbb{R})$,*

$$\int_{\mathbb{R}} \varphi d\nu = \varphi(-1) + \varphi(+1). \quad (12)$$

In particular, for every test function $\psi \in C_c(\mathbb{R})$ with $\psi(1) = 0$ and $\psi(-1) = 0$, $\int \psi d\nu = 0$, so $\nu|_{\mathbb{R} \setminus \{-1, +1\}} = 0$.

Proof. The first identity is a direct integration of φ against $\delta_{-1} + \delta_{+1}$. The second is the specialization to ψ vanishing at both atoms. $\square$

5 The pre-limit spectral density and its convergence to ν in $\mathcal{S}'(\mathbb{R})$

Proposition 9 (Pre-limit Bochner measure ν_X). *For every $X > X_0$, there exists a finite positive Borel measure ν_X on $\mathbb{R}$ satisfying*

$$C_X(\eta) = \int_{\mathbb{R}} e^{i\xi\eta} \nu_X(d\xi), \quad \eta \in [-(X - X_0), X - X_0], \quad (13)$$

given explicitly by

$$\nu_X = \frac{1}{X - X_0} |\widehat{\widetilde{H}}_X|^2(\xi) d\xi, \quad \widetilde{H}_X(x) := H(x) \mathbf{1}_{[X_0, X]}(x), \quad (14)$$

with $\widehat{\widetilde{H}}_X$ the Fourier transform of $\widetilde{H}_X \in L^2(\mathbb{R})$. In particular, ν_X is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}$, with L^1 density

$$\frac{d\nu_X}{d\xi} = \frac{|\widehat{\widetilde{H}}_X(\xi)|^2}{X - X_0}.$$

Proof. Since $\tilde{H}_X \in L^2(\mathbb{R})$ is real-valued with compact support $[X_0, X]$, Plancherel and the Wiener–Khinchin identity give, for every $\eta \in \mathbb{R}$,

$$\int_{\mathbb{R}} \tilde{H}_X(x) \overline{\tilde{H}_X(x+\eta)} dx = \int_{\mathbb{R}} e^{i\xi\eta} |\widehat{\tilde{H}_X}(\xi)|^2 d\xi,$$

where $\widehat{\tilde{H}_X}(\xi) = \int_{\mathbb{R}} \tilde{H}_X(x) e^{-i\xi x} dx$. Divide both sides by $X - X_0$ to obtain $C_X(\eta) = \int e^{i\xi\eta} \nu_X(d\xi)$ with ν_X as in (14). Positivity is from $|\widehat{\tilde{H}_X}|^2 \geq 0$; finiteness from

$$\nu_X(\mathbb{R}) = \frac{\|\widehat{\tilde{H}_X}\|_{L^2}^2}{X - X_0} = \frac{\|\tilde{H}_X\|_{L^2}^2}{X - X_0} = \frac{1}{X - X_0} \int_{X_0}^X |H(x)|^2 dx = C_X(0) < \infty.$$

The restriction $\eta \in [-(X - X_0), X - X_0]$ in (13) is the range of lags at which the windowed autocorrelation (1) is nonzero; outside it, both sides of (13) vanish by compact support of $\tilde{H}_X$. $\square$

Proposition 10 ($\nu_X \rightarrow \nu$ in $\mathcal{S}'(\mathbb{R})$). *For every $\varphi \in \mathcal{S}(\mathbb{R})$,*

$$\int_{\mathbb{R}} \varphi d\nu_X \xrightarrow{X \rightarrow \infty} \int_{\mathbb{R}} \varphi d\nu = \varphi(-1) + \varphi(+1). \quad (15)$$

Proof. Fix $\varphi \in \mathcal{S}(\mathbb{R})$. By the Fourier inversion formula on $\mathcal{S}$,

$$\varphi(\xi) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\varphi}(\eta) e^{i\xi\eta} d\eta, \quad \widehat{\varphi}(\eta) := \int_{\mathbb{R}} \varphi(\xi) e^{-i\xi\eta} d\xi.$$

Pair with ν_X and use Fubini (valid since $|\widehat{\varphi}(\eta)|$ is rapidly decreasing and ν_X is a finite measure):

$$\int_{\mathbb{R}} \varphi d\nu_X = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\varphi}(\eta) \int_{\mathbb{R}} e^{i\xi\eta} d\nu_X(\xi) d\eta = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\varphi}(\eta) C_X^\flat(\eta) d\eta,$$

where $C_X^\flat(\eta) := \int e^{i\xi\eta} d\nu_X(\xi)$ extends (13) to all $\eta \in \mathbb{R}$ via the Fourier-transform identity (both sides of (13) in fact coincide with the Fourier–Plancherel transform of $|\widehat{\tilde{H}_X}|^2$, a continuous bounded function on $\mathbb{R}$; and $C_X^\flat(\eta) = C_X(\eta)$ for $|\eta| \leq X - X_0$, $C_X^\flat(\eta) = 0$ otherwise by compact support of $\tilde{H}_X$).

Analogously, $\int \varphi d\nu = \frac{1}{2\pi} \int \widehat{\varphi}(\eta) C(\eta) d\eta = \frac{1}{2\pi} \int \widehat{\varphi}(\eta) \cdot 2 \cos |\eta| d\eta = \varphi(1) + \varphi(-1)$ by (11).

Therefore

$$\int \varphi d\nu_X - \int \varphi d\nu = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\varphi}(\eta) (C_X^\flat(\eta) - C(\eta)) d\eta. \quad (16)$$

Split the right side of (16) at $|\eta| = X - X_0$:

- For $|\eta| \leq X - X_0$: $C_X^\flat(\eta) = C_X(\eta)$, so $|C_X^\flat(\eta) - C(\eta)| = |C_X(\eta) - C(\eta)|$. By Theorem 1 and $C = \lim C_{\text{diag}}$,

$$|C_X(\eta) - C(\eta)| \leq |C_X(\eta) - C_{\text{diag}}(\eta)| + |C_{\text{diag}}(\eta) - C(\eta)| \leq \frac{8\pi(1 + |\eta|) \log^3(X/X_0)}{X - X_0} + \frac{0.349}{\sqrt{T_0}} + \varepsilon(\eta, X),$$

with $\varepsilon(\eta, X) \rightarrow 0$ as $X \rightarrow \infty$ uniformly on every compact η -set (Theorem 1, final sentence).

- For $|\eta| > X - X_0$: $C_X^\flat(\eta) = 0$ and $|C(\eta)| = |2 \cos |\eta|| \leq 2$.

In the first regime the contribution to |RHS of (16)| is bounded by

$$\frac{1}{2\pi} \int_{|\eta| \leq X-X_0} |\widehat{\varphi}(\eta)| \left(\frac{8\pi(1+|\eta|)\log^3(X/X_0)}{X-X_0} + \frac{0.349}{\sqrt{T_0}} + \varepsilon(\eta, X) \right) d\eta,$$

which tends to 0 as $X \rightarrow \infty$ by dominated convergence ($\widehat{\varphi} \in \mathcal{S}$ is integrable, $(1+|\eta|)\widehat{\varphi}(\eta)$ is integrable, and both $\log^3(X/X_0)/(X-X_0) \rightarrow 0$ and $\varepsilon(\eta, X) \rightarrow 0$ pointwise a.e.); the T_0 -dependent term contributes at most $0.349\|\widehat{\varphi}\|_{L^1}/(2\pi\sqrt{T_0})$ which is bounded uniformly in X for each fixed T_0 and sent to 0 in the final limit $T_0 \rightarrow \infty$ as in Theorem 1. In the second regime the contribution is bounded by

$$\frac{1}{2\pi} \int_{|\eta| > X-X_0} |\widehat{\varphi}(\eta)| \cdot 2 d\eta \leq \frac{1}{\pi} \int_{|\eta| > X-X_0} |\widehat{\varphi}(\eta)| d\eta \xrightarrow{X \rightarrow \infty} 0$$

by dominated convergence, since $\widehat{\varphi} \in \mathcal{S} \subset L^1$. Adding the two contributions proves (15). $\square$

Remark 11 (Interpretation). Proposition 10 states exactly what “existence of the spectral measure in the limit” means for the warped Hardy signal H : the pre-limit absolutely continuous spectral measure

$$\nu_X = \frac{|\widehat{H}_X(\xi)|^2}{X-X_0} d\xi$$

— whose density on $[-1, 1]$ is the periodogram of the windowed Hardy signal and whose total mass on $\mathbb{R}$ equals $C_X(0)$, the Hardy–Littlewood second-moment approximant — converges in the sense of tempered distributions to the two-atom measure $\nu = \delta_{-1} + \delta_{+1}$ as $X \rightarrow \infty$. The quantitative rate in Theorem 1 is the rate at which the continuous pre-limit spectral mass on $(-1, 1)$ is driven into the two band-edge points.

6 Corollaries

Corollary 12 (Total mass identity $\sigma_H^2 = 2$). $C(0) = \nu(\mathbb{R}) = 2$. *This matches the normalization $\sigma_H^2 = 2$ established via Hardy–Littlewood in `Appendix2GaussianStructure.tex`, Proposition 4.*

Proof. Directly from (4) at $\eta = 0$: $C(0) = 2 \cos 0 = 2$, and $\nu(\mathbb{R}) = C(0)$ from Theorem 5. $\square$

Corollary 13 (Band-edge vanishing of the off-band mass in the limit). *For every compact $K \subset \mathbb{R} \setminus \{-1, +1\}$,*

$$\nu(K) = 0. \tag{17}$$

Proof. From (10), $\nu = \delta_{-1} + \delta_{+1}$; the set K is disjoint from $\{-1, +1\}$, so $\nu(K) = \delta_{-1}(K) + \delta_{+1}(K) = 0 + 0 = 0$. $\square$

Corollary 14 (Equivalent pointwise form of the limit). *For every $\eta \in \mathbb{R}$,*

$$\lim_{X \rightarrow \infty} C_X(\eta) = \int_{\mathbb{R}} e^{i\xi\eta} \nu(d\xi) = e^{-i\eta} + e^{+i\eta} = 2 \cos |\eta|. \tag{18}$$

Proof. First equality is Theorem 1 combined with (9); the second is the explicit evaluation $\int e^{i\xi\eta} d(\delta_{-1} + \delta_{+1})(\xi)$; the third is $e^{i\eta} + e^{-i\eta} = 2 \cos \eta = 2 \cos |\eta|$. $\square$

7 Summary

1. The time-averaged covariance C_X of the warped Hardy signal H on $[X_0, X]$ is, for each X , the Fourier transform of a finite absolutely continuous positive measure ν_X on $\mathbb{R}$ with density equal to the normalized windowed periodogram of H (Proposition 9).
2. Theorem A of *Three Missing Theorems* gives $C_X(\eta) \rightarrow C(\eta) = 2 \cos |\eta|$ pointwise as $X \rightarrow \infty$, with explicit quantitative rate.
3. The limit C is continuous, Hermitian, and positive-definite (Lemmas 2, 4 and Corollary 3).
4. Bochner–Herglotz produces a unique finite positive Borel measure ν on $\mathbb{R}$ with $C = \widehat{\nu}$, and direct evaluation shows $\nu = \delta_{-1} + \delta_{+1}$ (Theorem 5).
5. The pre-limit Bochner measures ν_X converge to ν in the sense of tempered distributions (Proposition 10).

The spectral measure of H therefore *exists in the limit*, is concentrated entirely on the band edges $\{-1, +1\}$ of the theta-pullback frequency variable, and each edge carries mass exactly 1.